Derivatives, Alternatives & Portfolio Management

Forwards · Futures · Options · Swaps · Hedge Funds · Private Equity · CAPM · Portfolio Theory

Forwards & Futures

Pricing, margin, basis & hedging with index futures

DERIVATIVES: THE FOUR BUILDING BLOCKS

Forward

Custom OTC contract to buy/sell an asset at a fixed price on a future date. Credit risk accumulates — no daily cash flows.

Futures

Exchange-traded, standardised forward. Marked to market daily via margin account. Clearinghouse eliminates counterparty risk.

Options

Right (not obligation) to buy or sell at the strike price. Buyer pays a premium. Seller (writer) keeps premium regardless.

Swaps

OTC agreement to exchange periodic cash flows (e.g. fixed vs floating). Essentially a series of forward contracts. ISDA docs.

FORWARD PRICING: NO-ARBITRAGE MODEL

No-Arbitrage Logic

- If $F_0 > S_0(1+r)^T$ → cash-and-carry arbitrage: buy spot, sell futures
- If $F_0 < S_0(1+r)^T$ → reverse cash-and-carry: sell spot, buy futures
- Both are riskless profits → market forces eliminate them
- Result: one unique no-arbitrage forward price

No-Arbitrage Forward Price

$$F_0 = S_0 \times (1 + r)^T$$

With income/costs (dividends, storage)

$$F_0 = (S_0 + \text{PV costs} - \text{PV income}) \times (1+r)^T$$

Worked Example

S_0 (spot price)	$120.00
r (risk-free)	4.0% p.a.
T (time)	0.5 yr
Dividends	$2.00 (in 3M)
PV(dividends)	$1.98
F_0	= (120 − 1.98) × 1.02
F_0	**= $120.38**

FUTURES VS FORWARDS: KEY DIFFERENCES

Feature	Forwards	Futures
Trading venue	OTC — bilateral	Exchange (CME, ICE, Eurex)
Standardisation	Fully customised	Standardised contract specs
Settlement	At maturity only	Daily MTM → margin account
Credit risk	Full counterparty risk	Clearinghouse guarantees
Liquidity	Low; hard to exit early	High; close via offsetting trade
Margin required	None typically	Initial + variation margin
Price transparency	Private/bilateral	Public & continuous
Typical users	Corporates, banks	Speculators, hedgers, arbitrageurs

BASIS, CONTANGO & BACKWARDATION

Basis

- Basis = Spot price − Futures price
- Basis converges to zero at expiry
- Positive basis: spot > futures (backwardation)
- Negative basis: spot < futures (contango)
- Basis risk: hedge is imperfect if basis moves unexpectedly

Contango vs Backwardation

- Contango: F > S — carry cost exceeds convenience yield
- Common for: financial assets, gold, most metals
- Backwardation: F < S — strong convenience yield
- Common for: oil, agricultural commodities with supply concern
- Roll yield: negative in contango, positive in backwardation

Contango (carry > convenience)

$$F > S \times (1+r)^T \rightarrow \text{slope upward}$$

Backwardation (convenience > carry)

$$F < S \times (1+r)^T \rightarrow \text{slope downward}$$

Roll Yield

- When long futures and roll to next contract: gain if backwardation (buy cheap front, sell dear next), lose in contango
- Key driver of commodity futures returns alongside spot price change and collateral yield

HEDGING WITH FUTURES: EQUITY BETA MANAGEMENT

Number of futures contracts to adjust portfolio beta

$$N = (\beta^* - \beta P) / \beta f \times (\text{Portfolio MV} / \text{Futures price} \times \text{Multiplier})$$

Given Parameters

Portfolio MV	**$10,000,000**
Current beta (βP)	**1.90**
Target beta (β*)	**0.90**
Futures beta (βf)	**1.00**
Futures price	**5,000**
Multiplier	**$250**

Calculation

```
N = (0.90 – 1.90) / 1.00
      × (10,000,000 / (5,000 × 250))

N = –1.00 × (10,000,000 / 1,250,000)

N = –1.00 × 8.00  =  –8 contracts

→  SHORT 8 S&P 500 futures
```

Expected Return Impact (Rf = 2%, ERP = 7.5%)

	Before hedge	After hedge
E(Rp) = Rf + β × ERP	2% + 1.90×7.5% = 16.3%	2% + 0.90×7.5% = 9.3%

Options

Calls, puts, payoffs, put-call parity & strategies

OPTIONS: CALLS, PUTS & MONEYNESS

Call Option

- Buyer has RIGHT to BUY asset at strike K
- Seller (writer) has OBLIGATION to deliver
- Buyer pays premium C upfront
- Max loss for buyer = premium paid
- Payoff at expiry = $\max(S^T - K,\ 0)$

Put Option

- Buyer has RIGHT to SELL asset at strike K
- Seller has OBLIGATION to buy
- Buyer pays premium P upfront
- Max gain for buyer = K (if stock → 0)
- Payoff at expiry = $\max(K - S^T,\ 0)$

Moneyness	Call (right to buy)	Put (right to sell)
In-the-Money (ITM)	$S > K$	$S < K$
At-the-Money (ATM)	$S \approx K$	$S \approx K$
Out-of-the-Money(OTM)	$S < K$	$S > K$

WHAT DRIVES OPTION VALUE? (BLACK-SCHOLES INPUTS)

Input	Effect on CALL	Effect on PUT	Why
Stock price S ↑	↑ Value	↓ Value	Call moves deeper ITM; put further OTM
Strike price K ↑	↓ Value	↑ Value	Harder to profit on call; easier on put
Time to expiry T ↑	↑ Value	↑ Value	More time = more chance of large move
Volatility σ ↑	↑ Value	↑ Value	Both benefit from bigger potential swings
Risk-free rate r ↑	↑ Slightly	↓ Slightly	PV(K) falls → forward price rises → call benefits
Dividends D ↑	↓ Value	↑ Value	Ex-div drop reduces expected stock price

PUT-CALL PARITY

European options, no dividends

$$C + PV(K) = P + S_0 \quad \rightarrow \quad C - P = S_0 - PV(K)$$

Interpretation

- Long call + invest PV(K) = Long put + Long stock
- Both portfolios pay $\max(S^T, K)$ at expiry
- If violated → riskless arbitrage opportunity
- Applies to European options only

Synthetic Positions

- Synthetic long stock: long call + short put
- Synthetic long call: long put + stock − PV(K)
- Synthetic long put: long call − stock + PV(K)
- Used by traders to replicate and hedge positions

Numerical Example

Given: S = 100, K = 100, r = 5%, T = 1yr, Call = 8.00

Put = C + PV(K) − S = 8.00 + 100/1.05 − 100 = 8.00 + 95.24 − 100 = **3.24**

OPTIONS STRATEGIES

Covered Call

Long stock + Short call. Income strategy. Upside capped at K. Max loss = S_0 − premium (stock goes to 0).

Protective Put

Long stock + Long put. Insurance strategy. Downside floored at K. Costs premium but unlimited upside preserved.

Bull Call Spread

Long low-K call + Short high-K call. Reduces net premium paid. Profit capped between the two strikes.

Bear Put Spread

Long high-K put + Short low-K put. Lower-cost bearish position. Profit between the two put strikes.

Long Straddle

Long call + Long put at same K, same expiry. Profits from large move in either direction. High premium.

Collar

Long stock + Long put + Short call. Bounded corridor. Often structured near zero net premium (zero-cost collar).

Swaps

Interest rate swaps, currency swaps & CDS

INTEREST RATE SWAPS (IRS)

IRS Mechanics

- Two parties swap fixed vs floating interest payments on a notional
- No exchange of notional — only net cash flows settle
- Pay-fixed = synthetic short bond (duration decreases)
- Receive-fixed = synthetic long bond (duration increases)
- Most liquid derivative market globally (SOFR-based post-LIBOR)

Periodic net payment (fixed payer perspective)

$$\text{Net paid} = (\text{Fixed} - \text{Floating}) \times \text{Notional} \times \text{days}/360$$

Uses

- Convert fixed-rate debt to floating (save if rates fall)
- Extend/reduce portfolio duration without trading bonds

Cash Flow Example ($10M, 3%)

Period	Fixed paid	SOFR rcvd	Net
Q1	$75,000	$62,500	Pay $12,500
Q2	$75,000	$75,000	$0
Q3	$75,000	$87,500	Recv $12,500

Duration impact: pay-fixed shortens; receive-fixed lengthens

CURRENCY SWAPS & CREDIT DEFAULT SWAPS (CDS)

Currency Swap

- Exchange interest AND principal in two different currencies
- Notional IS exchanged at start and end (unlike IRS)
- Structures: fixed/fixed, fixed/float, float/float
- Used to hedge FX risk on foreign currency borrowing
- Cross-currency basis spread: deviation from covered interest parity

Credit Default Swap (CDS)

- Protection buyer pays periodic spread (CDS spread, in bps)
- Protection seller pays par on credit event (default, restructuring)
- Reference entity: a specific issuer or bond
- CDS spread ≈ bond credit spread (no-arbitrage link)
- Naked CDS: speculative — buyer has no underlying bond

CDS spread approximation

$$\text{CDS spread} \approx \text{PD} \times \text{LGD}$$

Basis: CDS vs bond spread

$$\text{Basis} = \text{CDS spread} - \text{Bond credit spread}$$

Total Return Swap (TRS)

- TRS payer pays total return of reference asset (income + capital gain/loss) and receives fixed or floating rate
- Gives synthetic leveraged exposure to an asset without owning it — widely used by hedge funds and CLO managers

Alternative Investments

Hedge funds, private equity, real assets & commodities

ALTERNATIVE INVESTMENTS: OVERVIEW

Hedge Funds

Pooled vehicles using long/short, leverage and derivatives targeting absolute return. High fees (2&20).

Private Equity

VC, growth equity, LBO. Illiquid 7–12yr fund life. J-curve cash flow profile. Higher return potential.

Real Estate

Direct property, REITs, private RE funds. Inflation hedge, income + capital appreciation. Cap rate metric.

Commodities

Energy, metals, agriculture via futures. Inflation hedge. Roll yield + convenience yield drive returns.

Infrastructure

Roads, airports, utilities. Inflation-linked regulated cash flows. Long duration, low correlation.

Private Credit

Direct lending, mezzanine, distressed. Illiquidity premium vs public credit. Growing institutional market.

HEDGE FUND STRATEGIES & STRUCTURE

Long/Short Equity	Long undervalued stocks, short overvalued. Net exposure adjustable. Most common HF strategy.
Global Macro	Directional bets on macro themes: currencies, rates, commodities (Soros/Druckenmiller style).
Event-Driven	M&A arbitrage, distressed debt, activist. Profit from corporate events regardless of market.
Relative Value / Arb	Exploit mispricings: fixed income arb, convertible arb, statistical arbitrage. Low net exposure.
CTA / Managed Futures	Systematic trend-following across asset classes. Uncorrelated to equity in crises.
Multi-Strategy	Allocate capital across sub-strategies dynamically. Internal capital reallocation improves Sharpe.

Fee structure: 2% management fee + 20% performance fee ('2-and-20'). High-water mark prevents double-dipping on recoveries.

PRIVATE EQUITY: VC, GROWTH & LBO

Venture Capital

- Seed through Series D+
- Early-stage; high failure rate
- Power law: few big winners fund the rest
- Returns via IPO or M&A exit
- Hold period: 3–7 years

Growth Equity

- Minority/majority in profitable co.
- Company has FCF; needs scale capital
- Less leverage than LBO
- Management stays in place
- Hold period: 3–6 years

LBO

- Acquire mature co. with 60–80% debt
- Stable FCF to service debt
- Value: ops improvement + deleveraging + multiple expansion
- PE equity: 20–40% of purchase price
- Hold period: 4–7 years

Return multiple

$$\text{MOIC} = \text{Exit Equity} / \text{Entry Equity}$$

Time-weighted return

$$\text{IRR: solves } \Sigma\ CFt / (1+IRR)^t = 0$$

J-Curve

- PE returns are negative early (management fees, slow capital deployment) then positive as companies are exited — the 'J' shape of cumulative net cash flows over fund life

PRIVATE EQUITY PERFORMANCE METRICS

MOIC / TVPI

2.5×

Total value / paid-in capital

IRR

22%

Internal rate of return

DPI

1.8×

Realised distributions / paid-in

Metric	Formula	What it measures
TVPI	(Distributions + Remaining Value) / Paid-in	> 1× = above water; headline multiple
DPI	Distributions / Paid-in Capital	Realised return — 'cash on cash'
RVPI	Residual (Unrealised) Value / Paid-in Capital	Paper gain, not yet crystallised
IRR	$\Sigma\ CFt / (1+IRR)^t = 0$	Time-sensitive; favours fast exits
PME	Compare PE IRR vs equivalent public index	Benchmark vs listed markets

REAL ASSETS: PROPERTY, INFRASTRUCTURE & COMMODITIES

Real Estate

- Direct: physical property (residential, commercial, industrial)
- REIT: exchange-traded; must distribute 90%+ of income
- Private RE fund: closed-end, commingled, illiquid
- Cap rate = Net Operating Income / Property Value

Infrastructure

- Roads, ports, airports, power grids, pipelines
- Regulated or contracted revenues; long concession life
- Inflation-linked pricing in many contracts
- Low correlation to equities; pension/insurance demand

Commodities (futures exposure)

- Returns = Spot return + Roll yield + Collateral yield
- Positive roll yield in backwardation, negative in contango
- Inflation hedge: strong historical correlation with CPI
- Low/negative correlation with bonds in inflationary periods

Timberland & Farmland

- Timberland: biological growth + land value appreciation
- Farmland: crop yield income + land appreciation
- Very low correlation to financial assets
- Both provide inflation sensitivity and diversification

Portfolio Theory & CAPM

Risk, return, diversification, efficient frontier & beta

PORTFOLIO RETURN & RISK: THE MATHS

Expected portfolio return

$$E(R_p) = \Sigma\, w_i \times E(R_i)$$

Portfolio variance (2 assets)

$$\sigma^2_p = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho\sigma_1\sigma_2$$

Correlation Effect on Risk

- $\rho = +1.0$: no diversification; risk is weighted average
- $\rho = 0$: partial diversification benefit
- $\rho = -1.0$: maximum diversification; risk can reach 0
- Key insight: portfolio risk < weighted avg whenever $\rho < 1$
- Increasing number of assets reduces unsystematic risk

Systematic vs Unsystematic Risk

- Total risk = Systematic + Unsystematic
- Systematic (market / beta risk): cannot be diversified away
- Unsystematic (idiosyncratic): eliminated by diversification
- Investors are compensated ONLY for systematic risk
- Therefore beta — not σ — determines expected return

Portfolio std deviation (2 assets)

$$\sigma_p = \sqrt{(w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2\cdot w_1\cdot w_2\cdot\rho\cdot\sigma_1\cdot\sigma_2)}$$

DIVERSIFICATION IN PRACTICE: BEACHWEAR VS DVD RENTAL

Classic CFA Example: Perfect Negative Correlation

Scenario	Probability	Beachwear Return	DVD Rental Return
Hot summer	50%	+25%	−5%
Cold summer	50%	−5%	+25%
Expected Return	**—**	**10%**	**10%**
Std Deviation	**—**	**15%**	**15%**

Portfolio (50% Beachwear + 50% DVD Rental):

Hot summer: 50%×25% + 50%×(−5%) = 10% Cold summer: 50%×(−5%) + 50%×25% = 10%

Portfolio σ = 0% (ρ = −1.0 perfectly offsets the returns)

EFFICIENT FRONTIER & CAPITAL ALLOCATION LINE (CAL)

Efficient Frontier

- Set of portfolios with maximum return for each level of risk
- Or minimum risk for each level of expected return
- Derived from Markowitz mean-variance optimisation
- No single 'best' portfolio without knowing preferences
- Dominated portfolios lie inside (below/right of) the frontier

Capital Allocation Line (CAL)

- Combines risk-free asset with risky (tangency) portfolio
- Tangency portfolio: point of tangency with the frontier
- Slope of CAL = Sharpe ratio of the tangency portfolio
- All investors choose a point on the CAL by their risk appetite
- With homogeneous expectations → CAL = CML (Capital Market Line)

Sharpe Ratio (= slope of CAL)

$$\text{Sharpe} = (E(R_p) - R_f) / \sigma_p$$

CML equation (along the line)

$$E(R_p) = R_f + [(E(R_m) - R_f)/\sigma_m] \times \sigma_p$$

Two-Fund Separation Theorem

- All investors hold the same risky portfolio (the tangency/market portfolio) and differ only in how much they allocate to the risk-free asset
- Conservative investor: more in Rf, less in market portfolio; Aggressive investor: leverages up the market portfolio

CAPITAL ASSET PRICING MODEL (CAPM)

CAPM Formula

$$E(R_i) = Rf + \beta_i \times [E(Rm) - Rf] = Rf + \beta_i \times ERP$$

Worked Example: Apple Inc.

Risk-free rate (Rf)	3.0%
Equity Risk Premium (ERP)	7.0%
Apple Beta (β)	1.50
E(R) = 3.0% + 1.5 × 7.0%	**= 13.5%**

CAPM Assumptions

- Mean-variance optimisers; homogeneous expectations
- Risk-free borrowing/lending; no taxes or transaction costs

Security Market Line (SML)

- Plots E(R) vs Beta for every asset
- Above SML → undervalued (positive alpha α)

CAPM Limitations

- Beta estimated from unstable historical data; Roll's critique: true market portfolio unobservable; single-factor model ignores size, value, momentum

BETA: MEASURING SYSTEMATIC RISK

Definition

$$\beta = Cov(R_i, Rm) / Var(Rm) = \rho(i,m) \times \sigma_i / \sigma m$$

Portfolio beta

$$\beta p = \Sigma\ w_i \times \beta_i$$

Beta	Example Asset	Market Sensitivity	E(R) at Rf=3%, ERP=7%
0.0	T-bill / cash	No market exposure	3.0% (risk-free)
0.5	Defensive utility stock	Moves 50% of market	6.5%
1.0	Index fund (by definition)	In line with market	10.0%
1.5	Apple Inc.	Amplified: 150% of market	13.5% ← Apple
1.9	High-beta cyclical	Very amplified market exposure	16.3% ← ANR before hedge
< 0	Inverse ETF / some gold	Moves opposite to market	Below Rf

ADJUSTING PORTFOLIO BETA WITH FUTURES: FULL EXAMPLE

Worked Example: Reducing Beta 1.90 → 0.90 (ANR-style)

Parameter	Value
Portfolio Market Value	$10,000,000
Current Beta (βP)	1.90
Target Beta (β*)	0.90
Futures Beta (βf)	1.00
Index Futures Price	5,000

	BEFORE	AFTER
Beta	1.90	0.90
Rf	2.0%	2.0%
ERP	7.5%	7.5%
E(R) = Rf+β×ERP	**16.3%**	**9.3%**

```
N = (β* − βP) / βf × (MV / F×Mult)
  = (0.90 − 1.90) / 1.00
      × (10,000,000 / (5,000×250))
  = −1.00 × (10,000,000/1,250,000)
  = −1.00 × 8.00
  = −8 contracts

→  SHORT 8 S&P 500 futures
```

By shorting 8 futures, beta drops 1.90→0.90.
E(R) falls from 16.3% to 9.3% without selling the portfolio.

Portfolio Construction

IPS, asset allocation, performance & risk management

INVESTMENT POLICY STATEMENT (IPS)

Return & Risk Objectives

- Return: required (liability-driven) or aspirational
- Risk tolerance: willingness + ability — both needed
- Time horizon: longer → higher risk capacity
- Liquidity: near-term needs reduce risk budget
- Tax: taxable vs tax-deferred accounts differ
- Legal & regulatory: govern permissible investments

Unique Circumstances

- ESG / ethical screening constraints
- Concentration limits (single stock, sector)
- Currency / geographic restrictions
- Employer stock or locked-up shares
- Derivatives or leverage restrictions
- SRI / impact investing mandates

Investor Type	Risk Tolerance	Time Horizon	Primary Objective
Young professional	High	30+ years	Wealth accumulation
Near-retiree	Low–Medium	5–10 years	Capital preservation + income
Endowment / Foundation	Medium–High	Perpetual	Spending rate + real return
Defined benefit pension	Depends on funding	Medium	Match liability growth (ALM)
Insurance company	Low	Short–Medium	Match claim reserves

STRATEGIC & TACTICAL ASSET ALLOCATION

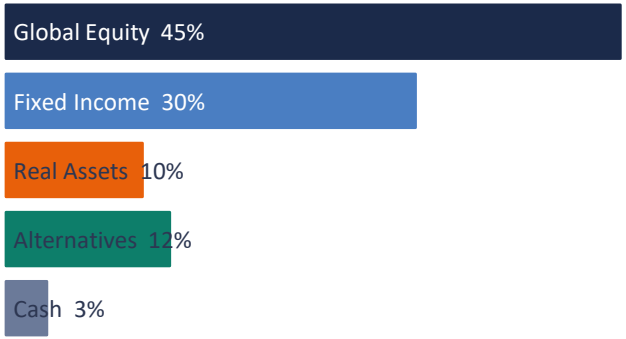
Strategic Asset Allocation (SAA)

- Long-term target weights based on IPS and CMAs
- Driven by capital market assumptions (returns, risks, correlations)
- Rebalancing: calendar-based or tolerance-band triggered
- Accounts for ~90% of return variability across portfolios
- Reviewed every 3–5 years or after major life change

Tactical Asset Allocation (TAA)

- Short-term deviations from SAA to exploit opportunities
- Based on valuations, macro views, momentum signals
- Adds alpha if timing is correct; costly if wrong
- Requires discipline to revert to SAA
- Separate overlay sleeve keeps SAA clean

Sample SAA



Rebalancing Strategies

- Calendar: rebalance quarterly / annually
- Tolerance band: trigger when weight drifts > X%
- Buy & hold: no rebalancing (maximum drift)
- CPPI: constant proportion portfolio insurance
- Risk parity: equal risk contribution from each asset

PORTFOLIO PERFORMANCE EVALUATION

Sharpe Ratio

$$(R_p - R_f)/\sigma_p$$

Return per unit of total risk

Treynor Ratio

$$(R_p - R_f)/\beta_p$$

Return per unit of beta risk

Jensen Alpha

$$R_p - CAPM$$

Excess vs CAPM prediction

Information Ratio

- IR = Active Return / Tracking Error
- Active return = Rp − Rb (vs benchmark)
- Tracking error = σ(Rp − Rb)
- IR > 0.50: skilled active manager
- IR > 1.00: exceptional skill

Attribution Analysis

- Brinson model: allocation + selection + interaction
- Allocation: over/under-weight sectors vs benchmark
- Selection: stock picks within sectors
- Goal: explain return relative to benchmark systematically
- Separates skill from luck and market exposure

RISK MANAGEMENT IN PORTFOLIO CONTEXT

Risk Measures

- Standard deviation: total volatility (appropriate for total portfolio)
- Beta: systematic market sensitivity
- VaR: max loss at confidence level over given horizon
- CVaR / ES: expected loss in the tail beyond VaR
- Max drawdown: worst peak-to-trough loss
- Tracking error: active risk vs benchmark

Value at Risk (VaR)

- VaR(95%, 1d): 5% chance of losing more than X in 1 day
- Parametric: assumes normal distribution
- Historical simulation: uses actual past returns
- Monte Carlo: simulates many random scenarios
- CVaR / ES solves VaR's tail-blindness limitation

Parametric VaR (normal distribution)

$$\text{VaR} = -(\mu - z \cdot \sigma) \cdot \sqrt{T} \cdot \text{Portfolio Value}$$

Risk Budgeting & Risk Parity

- Risk budgeting: allocate VaR or tracking error budget, not just capital
- Risk parity: each asset contributes equally to portfolio risk
- Both approaches better diversify risk exposure across assets
- Common use: multi-asset strategies, macro hedge funds

MULTI-FACTOR MODELS: BEYOND CAPM

Fama-French 3-Factor Model

- Market factor: Rm − Rf (market risk premium)
- Size factor (SMB): small-cap minus large-cap returns
- Value factor (HML): high B/P (value) minus low B/P (growth)
- R^2 improves from ~70% (CAPM) to ~90%
- Academic consensus: value and size premiums are persistent

Arbitrage Pricing Theory (APT)

- Multiple systematic macro factors
- Each has its own beta and risk premium λ
- No single unobservable market portfolio required
- Factors: GDP, inflation, credit spreads, oil price

Fama-French 3-Factor

$$E(R) = Rf + \beta m \cdot ERP + \beta s \cdot SMB + \beta v \cdot HML$$

APT general form

$$E(R) = Rf + \Sigma\ \beta k \times \lambda k$$

Carhart 4-Factor

- Adds momentum factor (MOM: past 12M winners minus losers)
- Momentum: prior-year return predicts next-month return

Smart Beta

- Systematic factor tilts: value, low-vol, quality
- Between passive index and active management

S10 · DERIVATIVES, ALTERNATIVES & PORTFOLIO — FORMULA REFERENCE

Forward price

$$F_0 = S_0 \times (1+r)^T$$

Beta-hedge (# contracts)

$$N = (\beta^* - \beta P)/\beta f \times MV/(F \times Mult)$$

Call payoff

$$\max(S^T - K,\ 0)$$

Put payoff

$$\max(K - S^T,\ 0)$$

Put-call parity

$$C + PV(K) = P + S_0$$

IRS net (fixed pays)

$$(Fixed - Float) \times Notional \times t/360$$

CAPM

$$E(R) = Rf + \beta \times ERP$$

Portfolio return

$$E(Rp) = \Sigma\ w_i \times E(R_i)$$

2-asset variance

$$\sigma^2 p = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho\sigma_1\sigma_2$$

Beta

$$\beta = Cov(R_i, Rm) / Var(Rm)$$

Sharpe ratio

$$Sharpe = (Rp - Rf) / \sigma p$$

Jensen alpha

$$\alpha = Rp - [Rf + \beta \cdot (Rm - Rf)]$$